GHULAM ISHAQ KHAN INSTITUTE OF ENGINEERING SCIENCES AND TECHNOLOGY

Assignment 2 – GPU Programming

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GitHub Repository: Link

Task 1: Initial Setup

Code: link

Output:

```
u2023559@pop-os:~/Assignment-2/task-1$ ./task01
Found 1 CUDA-capable GPU(s) on this system

=== GPU 0: "NVIDIA RTX A2000 12GB" ===

  [Architecture]
  Compute capability:                  8.6 — CUDA architecture version (major.minor)

  [Compute Resources]
  Streaming multiprocessors (SMs):     26   — Number of parallel processing units
  CUDA cores per SM:                   128  — Fixed for Ampere (GA106) architecture
  Warp size:                           32   — Threads executed in lockstep per warp
  Max threads per block:               1024   — Upper thread count limit per block
  Registers per block:                 65536   — Hardware registers available per block
  Core clock rate:                     1.20 GHz — GPU shader/core frequency

  [Thread Hierarchy Limits]
  Max block dimensions  (x, y, z):     1024 x 1024 x 64 — Per-dimension block size limits
  Max grid dimensions   (x, y, z):     2147483647 x 65535 x 65535 — Per-dimension grid size limits

  [Memory]
  Global memory:                       11.74 GB — Total device DRAM available
  Constant memory:                     64.00 KB — Read-only cache memory per device
  Shared memory per block:             48.00 KB — Fast on-chip memory per thread block
  L2 cache size:                       3.00 MB — Size of the device L2 cache
  Memory bus width:                    192-bit  — Width of the memory interface
  Memory clock rate:                   6001 MHz  — VRAM operating frequency

  [Derived Metrics]
  Peak memory bandwidth:               288.05 GB/s  — (2 x mem_clock x bus_width)
  Peak FP32 throughput:                7987.20 GFLOPS — (SMs x cores/SM x clock x 2)
u2023559@pop-os:~/Assignment-2/task-1$
```

Task 2:

Code: link

Output:

```
u2023559@pop-os:~/Assignment-2/task-2$ make
g++ -O3 -Wall -Wextra -std=c++17 -o matmul task2.cpp
u2023559@pop-os:~/Assignment-2/task-2$ ./matmul ../input2.txt output.txt
Output written to: output.txt
u2023559@pop-os:~/Assignment-2/task-2$ cat output.txt
# Result Matrix C: 3 x 2
rows_C=3
cols_C=2

1.2951 1.6336
1.2974 1.7321
0.7333 1.0618
u2023559@pop-os:~/Assignment-2/task-2$
```

Task 3:

Code: link

Output:

```
u2023559@pop-os:~/Assignment-2/task-3$ make
nvcc -O3 -Xcompiler -Wall,-Wextra -arch=sm_86 -o matmul_gpu task3.cu
u2023559@pop-os:~/Assignment-2/task-3$ ./matmul_gpu ../input2.txt output.txt
Output written to: output.txt
Computational Intensity: 0.4615 FLOPS/byte
u2023559@pop-os:~/Assignment-2/task-3$ cat output.txt
# Computational Intensity (Naive GPU)
#   FLOPS      = 2 * N * M * K = 2 * 3 * 2 * 4 = 48
#   Mem traffic = (N*K + K*M + N*M) * 4 bytes
#   CI          = 0.4615 FLOPS/byte
#
# Result Matrix C: 3 x 2
rows_C=3
cols_C=2

1.2951 1.6336
1.2974 1.7321
0.7333 1.0618
u2023559@pop-os:~/Assignment-2/task-3$
```

Task 4:

Code: link

Output:

```
u2023559@pop-os:~/Assignment-2$ g++ -O3 -Wall -o data_generator data_generator.cpp
./data_generator 64   64   64   input_64.txt
./data_generator 128  128  128  input_128.txt
./data_generator 256  256  256  input_256.txt
./data_generator 512  512  512  input_512.txt
./data_generator 1024 1024 1024 input_1024.txt
./data_generator 2048 2048 2048 input_2048.txt
Generated: input_64.txt  (A: 64x64, B: 64x64)
Generated: input_128.txt  (A: 128x128, B: 128x128)
Generated: input_256.txt  (A: 256x256, B: 256x256)
Generated: input_512.txt  (A: 512x512, B: 512x512)
Generated: input_1024.txt  (A: 1024x1024, B: 1024x1024)
Generated: input_2048.txt  (A: 2048x2048, B: 2048x2048)
u2023559@pop-os:~/Assignment-2$ cd task-4
u2023559@pop-os:~/Assignment-2/task-4$ make
g++ -O3 -Wall -Wextra -std=c++17 -o matmul_cpu task4_cpu.cpp
nvcc -O3 -Xcompiler -Wall,-Wextra -arch=sm_86 -o matmul_gpu task4_gpu.cu
u2023559@pop-os:~/Assignment-2/task-4$ ./matmul_cpu ../input_64.txt outputc_64
./matmul_cpu ../input_128.txt outputc_128
./matmul_cpu ../input_256.txt outputc_256
./matmul_cpu ../input_512.txt outputc_512
./matmul_cpu ../input_1024.txt outputc_1024
./matmul_cpu ../input_2048.txt outputc_2048
Size: 64x64x64  CPU Time: 586 us
Size: 128x128x128  CPU Time: 4597 us
Size: 256x256x256  CPU Time: 16476 us
Size: 512x512x512  CPU Time: 166515 us
Size: 1024x1024x1024  CPU Time: 2.97787e+06 us
Size: 2048x2048x2048  CPU Time: 3.12984e+07 us
u2023559@pop-os:~/Assignment-2/task-4$ ./matmul_gpu ../input_64.txt outputg_64
./matmul_gpu ../input_128.txt outputg_128
./matmul_gpu ../input_256.txt outputg_256
./matmul_gpu ../input_512.txt outputg_512
./matmul_gpu ../input_1024.txt outputg_1024
./matmul_gpu ../input_2048.txt outputg_2048
Size: 64x64x64  GPU Time (H2D+kernel+D2H): 76.928 us
Size: 128x128x128  GPU Time (H2D+kernel+D2H): 129.504 us
Size: 256x256x256  GPU Time (H2D+kernel+D2H): 397.76 us
Size: 512x512x512  GPU Time (H2D+kernel+D2H): 1704.16 us
Size: 1024x1024x1024  GPU Time (H2D+kernel+D2H): 8155.78 us
Size: 2048x2048x2048  GPU Time (H2D+kernel+D2H): 50784.8 us
u2023559@pop-os:~/Assignment-2/task-4$
```

Task 5:

Code: link

Output:

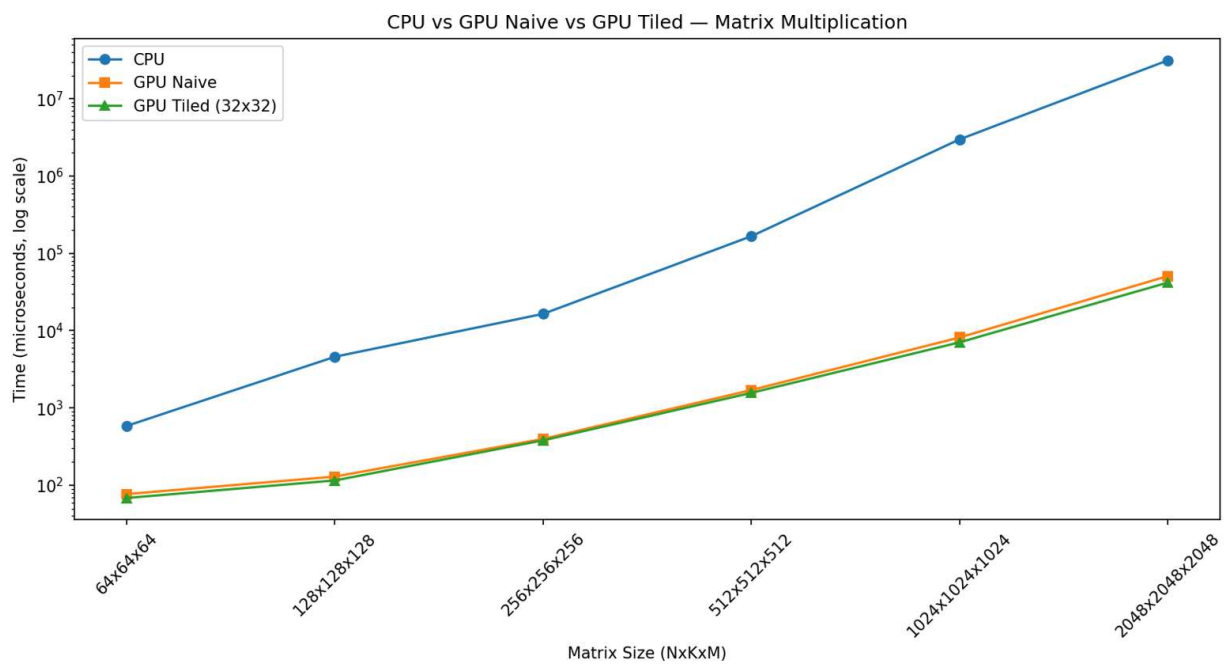
```
u2023559@pop-os:~/Assignment-2/task-5$ make
nvcc -O3 -Xcompiler -Wall,-Wextra -arch=sm_86 -o tiletest task5_tiletest.cu
nvcc -O3 -Xcompiler -Wall,-Wextra -arch=sm_86 -o matmul_tiled task5_tiled.cu
u2023559@pop-os:~/Assignment-2/task-5$ ./tiletest ../input_1024.txt
Testing tile sizes on 1024x1024 x 1024x1024

Tile Size  8x8  : 7739.52 us  (shared mem: 512 bytes)
Tile Size 16x16 : 6572.16 us  (shared mem: 2048 bytes)
Tile Size 32x32 : 5566.02 us  (shared mem: 8192 bytes)
(Shared memory limit on RTX A2000: 49152 bytes = 48 KB)
(Max threads/block = 1024, so TILE > 32 is not possible)

=> Fastest tile size: 32x32 (5566.02 us)

Tile timings written to tile_timings.csv
u2023559@pop-os:~/Assignment-2/task-5$ ./matmul_tiled ../input_64.txt outputt_64.txt
./matmul_tiled ../input_128.txt outputt_128.txt
./matmul_tiled ../input_256.txt outputt_256.txt
./matmul_tiled ../input_512.txt outputt_512.txt
./matmul_tiled ../input_1024.txt outputt_1024.txt
./matmul_tiled ../input_2048.txt outputt_2048.txt
Size: 64x64x64  GPU Tiled Time (H2D+kernel+D2H): 103.2 us
Output written to: outputt_64.txt
CI (naive): 10.6667 FLOPS/byte
CI (tiled): 341.3333 FLOPS/byte
Size: 128x128x128  GPU Tiled Time (H2D+kernel+D2H): 118.016 us
Output written to: outputt_128.txt
CI (naive): 21.3333 FLOPS/byte
CI (tiled): 682.6667 FLOPS/byte
Size: 256x256x256  GPU Tiled Time (H2D+kernel+D2H): 390.656 us
Output written to: outputt_256.txt
CI (naive): 42.6667 FLOPS/byte
CI (tiled): 1365.3333 FLOPS/byte
Size: 512x512x512  GPU Tiled Time (H2D+kernel+D2H): 1608.06 us
Output written to: outputt_512.txt
CI (naive): 85.3333 FLOPS/byte
CI (tiled): 2730.6667 FLOPS/byte
Size: 1024x1024x1024  GPU Tiled Time (H2D+kernel+D2H): 7311.04 us
Output written to: outputt_1024.txt
CI (naive): 170.6667 FLOPS/byte
CI (tiled): 5461.3333 FLOPS/byte
Size: 2048x2048x2048  GPU Tiled Time (H2D+kernel+D2H): 41706.5 us
Output written to: outputt_2048.txt
CI (naive): 341.3333 FLOPS/byte
CI (tiled): 10922.6667 FLOPS/byte
u2023559@pop-os:~/Assignment-2/task-5$ python3 plot.py
Plot saved to comparison.png
u2023559@pop-os:~/Assignment-2/task-5$ python3 plot_tile_size.py
Plot saved to best_tile_size.png
```

Input.txt format:

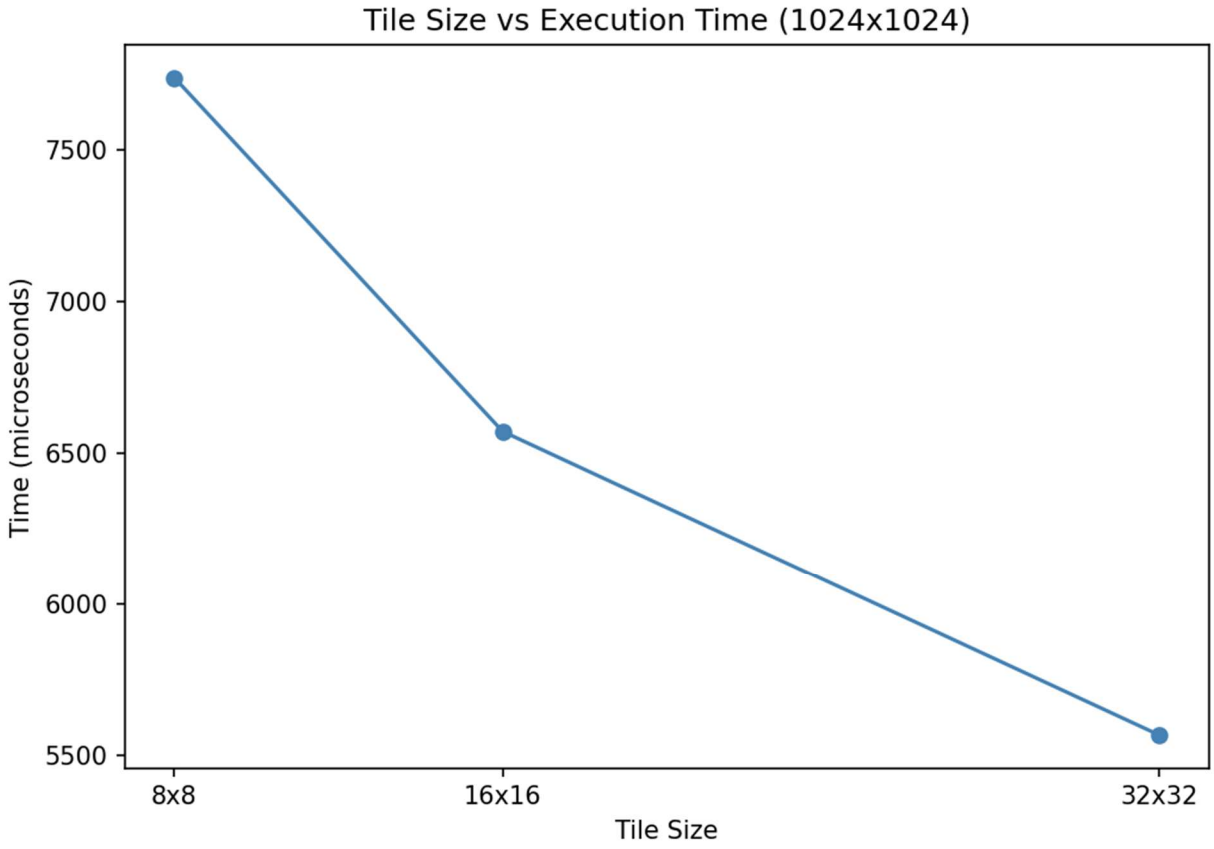
```
# Matrix A: N x K
rows_A=N
cols_A=K

<N rows of K space-separated floats>

# Matrix B: K x M
rows_B=K
cols_B=M
```

- I have used Makefile as a build system for compiling the programs

Plots:



GPU Tiled clearly outperforms naïve implementation on GPU and CPU implementation.



32 is the maximum best size for the tile, for it gives the least time. (And we can't go beyond 32 tile size because of our GPU's shared memory size limit).

Ends Here